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Impact of economic policy uncertainty on exchange rate volatility of China

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ABSTRACT

This study investigates the impact of economic policy uncertainty (EPU) on China's exchange rate volatility from December 2001 to November 2018. Using the quantile regression, our results show that the impact of EPU on exchange rate volatility in China exhibits asymmetry as well as heterogeneity in different markets. The EPU for China impacts positively and significantly on all quantiles volatilities of exchange rates. Furthermore, we observe that EPU has a mixed effect on exchange rate volatility with apparent economy-by-economy differences. The US, Europe and Japan EPU have significant impacts, while Hong Kong EPU is insignificantly correlated with exchange rate volatility.

1. Introduction

The ongoing study on the link between economic policy uncertainty (EPU) and exchange rate volatility has not received attention, even though extensive literatures explore the causes of exchange rate volatility or the heterogeneous effects of EPU. On the one hand, EPU exerts important effects on asset prices in different markets, including stock, future, commodities and insurance (Dong et al., 2019; Fang et al., 2018; Gupta et al., 2018; Li et al., 2018; Phan et al., 2018; Shahzad et al., 2017). On the other hand, scholars also explore the influence factors on exchange rate volatility in international economics (Aftab et al., 2015; Wang and Morley, 2018; Vithessonthi, 2014). However, there is rare literature about the impact of EPU on exchange rate volatility (Balcilar et al., 2016; Beckmann and Czudaj, 2017). Theoretically, EPU has both direct and indirect impacts on exchange rate volatility. The greater EPU a country has, the faster exchange rates will change with a more flexible regime. Moreover, EPU influences the economy state, thereby impacting exchange rate volatility. Therefore, we explore the impact of EPU on exchange rate volatility.

EPU is a definite factor of asymmetric exchange rate volatility. It is well-known that EPU would have a positive effect on the volatility of exchange rates. This result has been primarily linked to government regulation (Brogaard and Detzel, 2015). Policies adopted by the central bank under different exchange rate fluctuation levels will lead to different impacts of EPU on exchange rates. Conversely, the effect of EPU shock on the correlations is weak during periods of drastic exchange rate volatility. When exchange rate volatility is relatively high, the effect is responded to changing economic circumstances and shocks through changes in great events. Thus, EPU may induce different and asymmetric shocks at various quantiles of exchange rate volatility.

There are heterogeneous effects of EPU in different markets on exchange rate volatility of China. The impact of EPU in different

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markets on the exchange rate volatility is mixed. Substantial effects of exchange rates may be due to the economic regime in different markets (Lee et al., 2019). For example, the effect is insignificant in Hong Kong. Besides, evidence also indicates that the degree of exchange rate volatility might notably be attributed to investors' decisions between China and other markets (Hu et al., 2018; Liao et al., 2019; Bao et al., 2018). Specifically, the change of EPU could impact various transmission mechanism, thereby bringing about the heterogeneous role on exchange rate volatility.

Our aim in this paper is to reveal the extent to which the relationship between EPU and exchange rate volatility on different quantiles with the quantile regression approach. This approach allows us to better assess the differences in the EPU influence across exchange rate volatility distribution in order to shed light on the impact of EPU upon exchange rate volatility asymmetry in China. Furthermore, we provide evidence on the heterogeneous effect in different markets.

This paper is structured as follows. Section 2 briefly introduces the analysis of data and methods. In Section 3, we test the asymmetry between the magnitude of exchange rate volatility and elaborate on EPU from different markets with regard to regional heterogeneity. Section 4 concludes the paper.

2. Methodology and data

2.1. Methodology

Most of the traditional methods on the impact of exchange rate volatility, only provides the conditional expectation of exchange rate volatility, fails to describe the whole picture of the conditional distribution. However, due to the tremendous heterogeneity of development in the China, the relationships between EPU and exchange rate volatility are likely to perform discriminately at different quantiles. In this regard, the quantile regression method allows the coefficients to vary with multiple quantiles, and it has distinctive advantages of detecting the variation in the effect of EPU on the distribution of exchange rate volatility. This prompts us to employ the quantile regression (QR).

Therefore, this paper adopts the quantile regression method to examine whether there are asymmetric effects of EPU on the conditional distribution of exchange rate volatility. Additionally, considering the overall international economic environment, we take into account heterogeneity to investigate the effects of EPU of different markets on exchange rate volatility. The conditional quantile model can be represented as (1)

$$Q_{\tau}(VOL|EPU_i) = \alpha_0^{\tau} + \beta^{\tau}EPU_i + \gamma^{\tau}X + \varepsilon^{\tau} \quad (1)$$

where $Q_{\tau}(VOL|\cdot)$ denotes a conditional quantile function of exchange rate volatility evaluated at τ th quantile, where $\tau \in (0, 1)$. VOL is the monthly exchange rate volatility. EPU_i represents economic policy uncertainty of different markets, including China (CN), Hong Kong (HK), the United States (US), Europe (EU) and Japan (JP). The coefficient vector β^{τ} measures the dependence degree of exchange rate volatility at τ th quantile to EPU_i , which is the focus of this study. X represents a set of control variables. ε^{τ} will differ depending on the particular quantile being estimated.

2.2. The data

Exchange rate volatility in China is considered in this model. Exchange rates of China are the bid–ask mid-prices of the yuan against the dollar, which could better depict the influence of exchange rates in the international market. Theoretically, the equilibrium exchange rate is valid in the long run in the international market, and there is not an arbitrage opportunity between different currencies (Kreps, 1981; Ross, 1976). The dollar peg also serves as an anchor for monetary policy in China (Tervala, 2019). The U.S. dollar is considered as exchange rates currency because it is used as the invoicing currency in international trading, the most important reserve currency in the world, and the currency in which the most international commercial transactions are made. Moreover, the choice of the USD/RMB exchange rates is also highlighted by the fact that China in flexibility exchange rate regime is very sensitive to changes in the US. Thus, we select monthly exchange rates in the direct quotation extracted from China Stock Market & Accounting Research Database (CSMAR) to measure exchange rate volatility during the period from January 2002 to November 2018. Further, we model the volatility of exchange rate based on the ARMA-ARCH specification in this paper¹ (Takaishi, 2018).

In addition, a key part of our empirical approach is to introduce a measure of EPU as a determinant of exchange rate volatility. We rely on data for compiling EPU index provided by Baker et al. (2016) at their companion website (<http://www.policyuncertainty.com/>) for a sample period between 2002 and 2018 at the monthly frequency. The index is a news-based index, which captures a wide range of policy uncertainty terms appeared in the countries newspapers. Thence, we focus on EPU factors as explanatory variables. We also address the role played by bilateral trade in determining exchange rate volatility. Our sample includes China (CN) and economies which are four major trading partners with China, including Hong Kong (HK), the United States (US), Europe (EU) and Japan (JP). The selected partner economies represent 67% of the value of foreign trade with China. China's major foreign trade is concentrated in the four trade relationships. This index is accessed from the World Trade Organization (<https://www.wto.org/>). All the original data are processed by taking natural logarithms to correct for potential heteroscedasticity and dimensional difference between series.

General macroeconomic factors must be controlled in any attempt to discuss the impact of EPU on exchange rate volatility. Firstly,

¹ Considering the space limit, we omit this part in this article.

Table 1
Effects of EPU in different markets^a.

	Dependent Variable: USD/RMB EXCHANGE RATE VOLATILITY				
	5%	25%	50%	75%	95%
CN	0.083*** (0.024)	0.315*** (0.060)	0.620*** (0.122)	1.033*** (0.253)	0.960* (0.516)
HK	0.010 (0.017)	0.023 (0.032)	0.079 (0.107)	0.028 (0.340)	−0.122 (0.741)
US	0.043 (0.036)	0.133** (0.060)	0.335* (0.177)	1.353** (0.543)	1.639 (1.188)
EU	0.028 (0.021)	0.126*** (0.042)	0.486*** (0.183)	1.681*** (0.418)	1.312* (0.783)
JP	−0.018 (0.034)	0.165** (0.067)	0.444* (0.242)	0.131 (0.500)	−1.780* (1.066)

Note: Numbers in parentheses and square brackets are standard errors and p-values, respectively. Significance levels: *10%, **5%, ***1%.

^a Considering the space limit, we omit additional details concerning the control variables in different markets.

exchange rate volatility is greatly influenced by capital flows in an open economy (Gelman et al., 2015). For one thing, an increase in the terms of trade (TOT) is the ratio between the price of exports and the price of imports that reflects an increase in export prices and reduces foreign export demand (Raza and Afshan, 2017). For another, an important factor that has a negative impact on exchange rate volatility is foreign exchange reserves (RES) in China (Eichler and Littke, 2018). RES is the log of China's foreign exchange reserves. Both of them are constructed from the CSMAR database. However, monetary policy has also been put forward as factors affecting exchange rates of developing countries (Mueller et al., 2017). Changes in monetary policy lead to significant volatility of exchange rates. We use the repurchase rate (INR) of the People's Bank of China, which is its key policy rate, as a proxy for China's monetary policy (Özmen and Yilmaz, 2017). In addition, considering the data frequency, we adopt the producer price index (PPI) which is measured by the percentage change year-on-year and compiled by the National Bureau of Statistics of China (Nusair and Kisswani, 2015). This is intended as a general check for potential omitted variable bias, since exchange rate volatility may systematically vary with the level of development, and it may just proxy for other economic developments associated with inflation.

3. Results

3.1. Asymmetry effect in China

In this section, we present quantile regression estimates of the effect of EPU in different markets on China's exchange rate volatility. We estimate the model at the 5%, 25%, 50%, 75%, and 95% quantiles. The results are presented in Table 1.

The impact of China EPU on exchange rate volatility at different levels is asymmetric. The results, reported in Table 1, show that EPU is positive and significant at all volatility of exchange rates. With the higher volatility of exchange rates, there is a greater positive shock of EPU, and the highest correlation for the 75% quantiles is observed between the volatility of exchange rate and EPU. However, the correlation is significantly low at drastic volatility. Governments are more likely to change economic policies when the volatility is increasing. Due to stability preference, the government would intervene to ensure changes in exchange rates during higher volatility. Therefore, EPU reflects decision making in the government which should be enhanced during higher volatility of exchange rates. What's more, there have been major economic and political reforms in recent decades. China's exchange rate system has been evolving gradually but constantly over the sample period. It is also inevitable that exchange rates of China experienced the recent US–China Trade Dispute. And certainty shocks, which brought by these events, cause the volatility of exchange rates. Therefore, it is natural to predict that the impact of EPU will be mitigated when the volatility of exchange rates is drastic.

3.2. Heterogeneity from different markets

Conditional quantile estimates for the regional heterogeneity are also explored by performing QR. The empirical results in Table 1 show heterogeneous outcomes, and the specific variations of coefficients across quantiles are depicted in Fig. 1. In particular, one can distinguish markets into two groups. The first one is Hong Kong where one can find an independent relationship between its EPU and exchange rate volatility. The second group includes the US, Europe and Japan where strongly significant effects linkage between exchange rate volatility and EPU is revealed.

Regarding the first group, the effects of Hong Kong EPU on exchange rate volatility in China are insignificant across the quantiles. In other words, EPU does not cause exchange rate volatility of China irrespective of whether exchange rate volatility is high or low. This makes sense since the economic system has a special situation between China and Hong Kong. Since 1997 reunification, a prominent aspect has been that the Chinese central government recentralizes divergent regulatory measures, which is the one-country two-systems (OCTS) mode, in order to govern Hong Kong. Even though Hong Kong retains previous different systems, it has been a fundamental part of the government system of mainland China. Thus, Hong Kong EPU has no significant impact on exchange rate volatility of China.

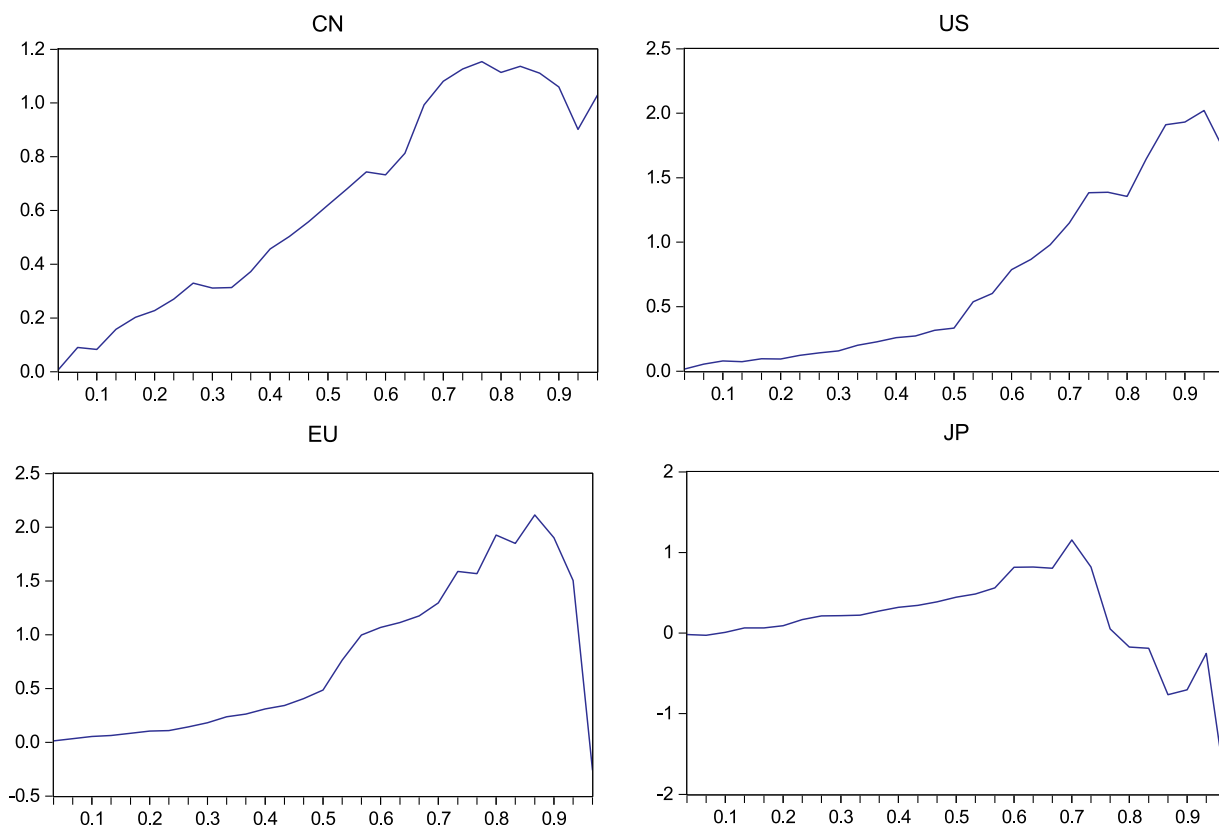


Fig. 1. Estimation results of quantile regression. ²

The impact of EPU on exchange rate volatility is significant in the second group. The effect of the US EPU is not negligible for all quantiles, except for the quantiles 5% and 95%, which implies the cross-border linkage between US EPU and China exchange rate volatility. And the impact significantly increases with the rise of exchange rates volatility. The US economy's poor performance has been caused by an exogenous increase in the US policy uncertainty. Considering changes in the international financial market brought about by the US economic state shocks, Chinese investors may well come up with new trading decisions accordingly. Therefore, changes in the US EPU is likely to influence the behavior of all investors who partake in the Chinese exchange rate markets. The positive and significant dependence between the Europe EPU and exchange rate volatility in China is evident for all quantiles, except for the quantile 5%. Thus, there is strong evidence that exchange rate volatility in China is the most sensitive to the Europe EPU. In addition, the impact of EPU on exchange rate volatility is an inverted U-shaped curve. The result suggests that the bigger the volatility of exchange rates is, the more sensitive bilateral trade brought by the EPU change so that changes in bilateral trade can increase market volatility. And the impact is similar across China for the 95% quantile, shocks in events mitigate EPU influence. In terms of Japan, the effects of EPU on exchange rate volatility in China are similar to that of the US and Europe, with dependence remaining almost increase across 25% and 50% quantiles. But Japan EPU seems to be not related to exchange rates in the 75% quantile, and the impact is negative in the upper quantiles. In fact, the impact of Japan EPU is consistent with the Europe when the volatility is normal, but there is a special influence during periods of higher volatility. EPU for Japan influences its economic state, provoking cooperation between Japan and China during the recent US-China Trade Dispute. This relation of trading leads to avoiding drastic volatility of exchange rates by increasing investors' confidence. Overall, this means that the dependence with respect to changes in the EPU is asymmetric for these markets.

4. Conclusions and policy implications

This paper implements an exhaustive QR analysis to investigate the relationship between exchange rate volatility in China and EPU of different markets. Our main findings can be summarized as follows: (i) There is an asymmetric impact of EPU on exchange rate volatility in China. EPU has a positive and significant impact at all quantiles, and there is an inverted-U relationship. (ii) There exist heterogeneous effects of EPU from different markets on exchange rate volatility across different quantiles, as well. More specifically, EPU of Hong Kong has no impacts on volatility. For the United States, the degree of the impact rises with the increase of

² we omit additional details concerning the effect of Hong Kong and the control variables in different markets.

quantiles, while that of Europe and Japan has a significant and inverted U-shaped effect on exchange rate volatility in China.

Policymakers must pick sound economic policies to promote prosperity. The clarity in economic policymaking, especially in the period of high volatility, can lead to more stable markets. Additionally, our findings are also of great relevance for policymakers on managing exchange rate fluctuations and on prevention of potential risks that may arise due to significant dependence among different markets. They should further control over risks at markets of significant dependence and properly decide on the timing and extent of foreign exchange rate intervention.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.frl.2019.08.014](https://doi.org/10.1016/j.frl.2019.08.014).

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